

ThermoTwin-F

Gas-Turbine Performance Simulator - Engineering Report

Model	Open/simple Brayton cycle, single-shaft
Working fluid	Air-standard with separate hot-gas properties
Analyses	Design - off-design - degradation - transient - uncertainty - diagnostics
Implementation	Modern Fortran (2008), modular, unit-tested
Post-processing	Python / matplotlib
Generated	2026-06-07

Scope and honesty note
This is a non-proprietary educational simulator. It uses standard textbook thermodynamics and representative property values. It is NOT calibrated or validated against any specific commercial engine, and it is NOT compliant with ASME PTC 22 or ISO 2314. The verification self-test checks the code against an independent hand calculation; it does not constitute validation against hardware.

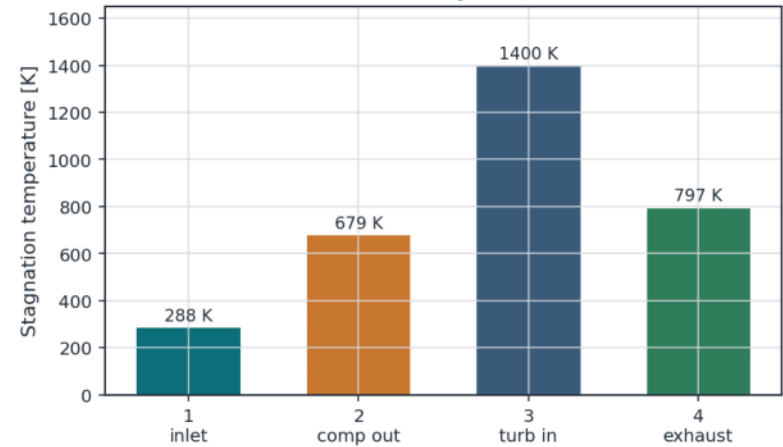
Design-point results summary

Quantity	Value	Unit
Compressor outlet T2	679.4	K
Turbine inlet T3 (TIT)	1400.0	K
Exhaust T4	797.1	K
Pressure ratio	15.0	-
Fuel-air ratio	0.0195	-
Air mass flow	100.0	kg/s
Compressor power	39.31	MW
Turbine power	70.57	MW
Net electrical power	29.87	MW
Heat input (fuel)	97.56	MW
Thermal efficiency	30.62	%
Heat rate	11756	kJ/kWh
Specific power	298.7	kW/(kg/s)
Exhaust energy	59.56	MW

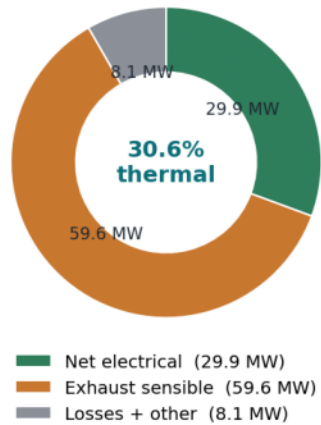
Design-point summary

ThermoTwin-F - Design-point summary

Station temperatures



Fuel energy disposition (input 97.6 MW)

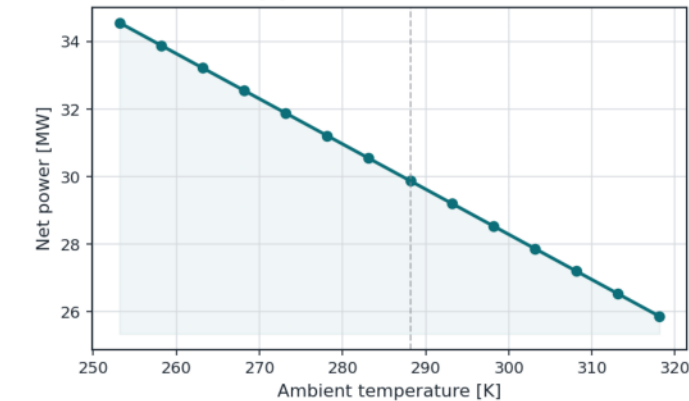


ThermoTwin-F - educational model, not validated against a specific engine

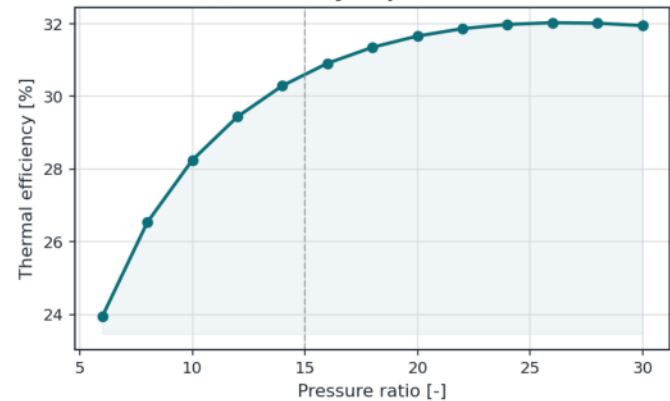
Parametric sensitivity

ThermoTwin-F - Parametric sensitivity

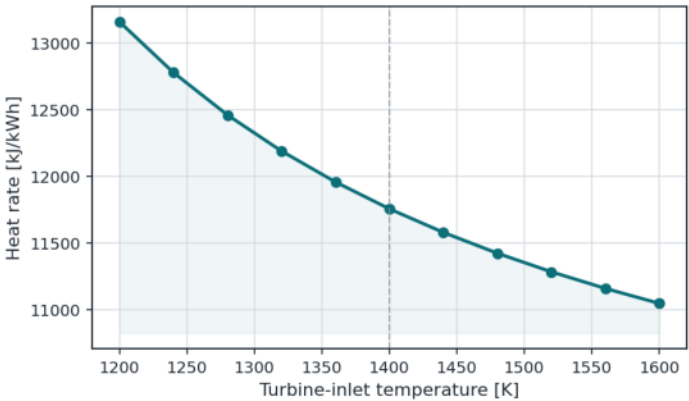
(a) Power vs ambient T



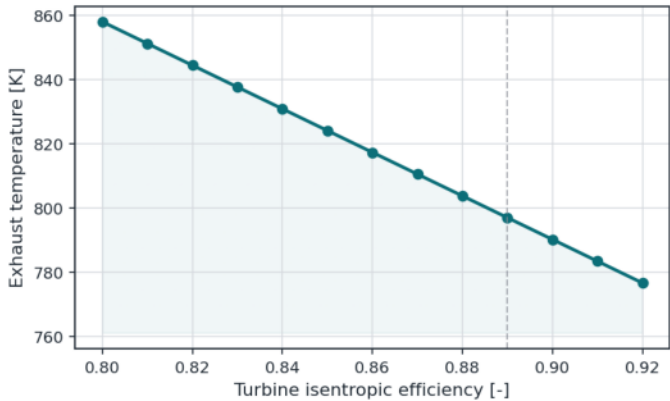
(b) Efficiency vs pressure ratio



(c) Heat rate vs TIT



(d) Exhaust T vs turbine eff.

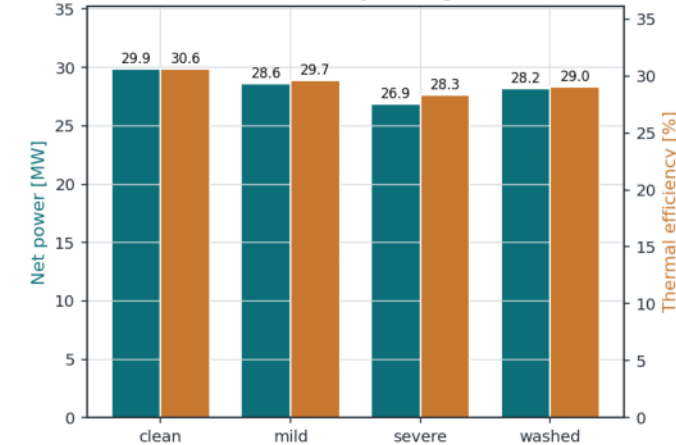


ThermoTwin-F - educational model, not validated against a specific engine

Degradation impact

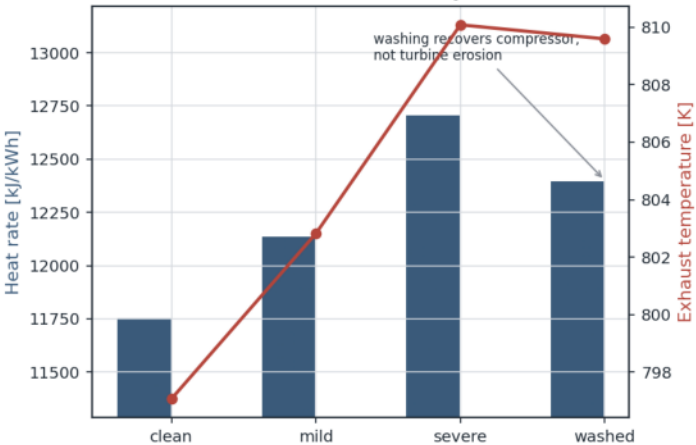
ThermoTwin-F - Degradation impact

Power & efficiency vs degradation



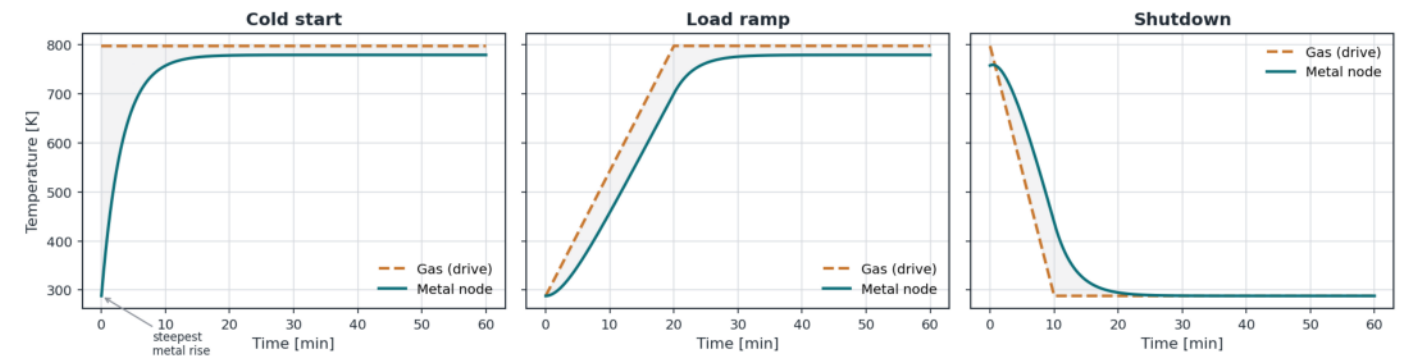
ThermoTwin-F - educational model, not validated against a specific engine

Heat rate & exhaust temperature



Transient thermal response

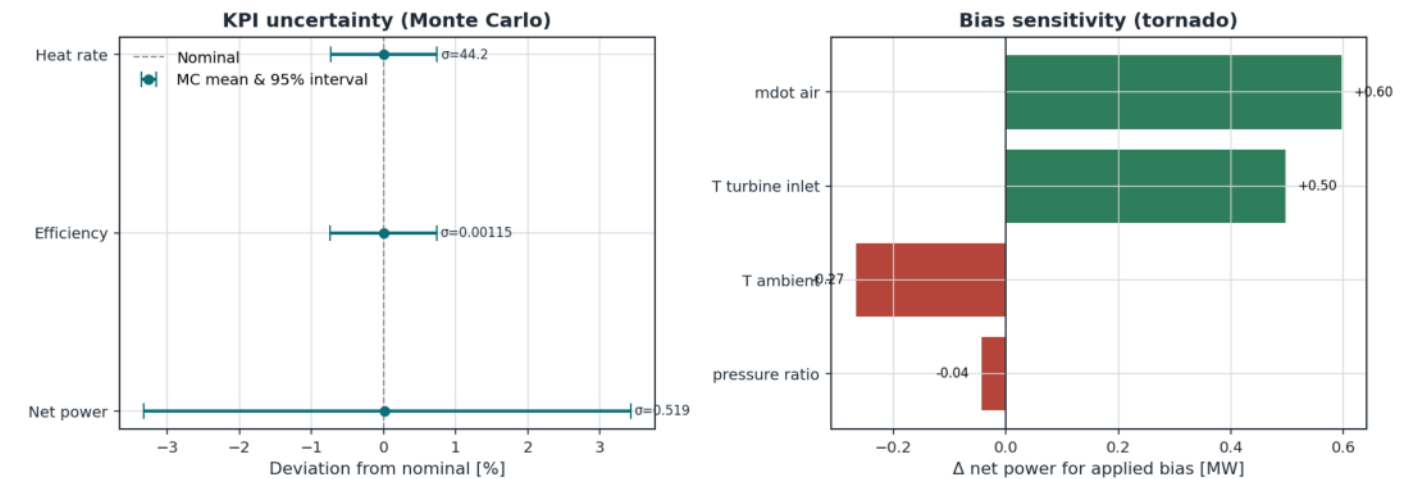
ThermoTwin-F - Transient hot-section thermal response (lumped node)



ThermoTwin-F - educational model, not validated against a specific engine

Measurement uncertainty

ThermoTwin-F - Measurement uncertainty propagation

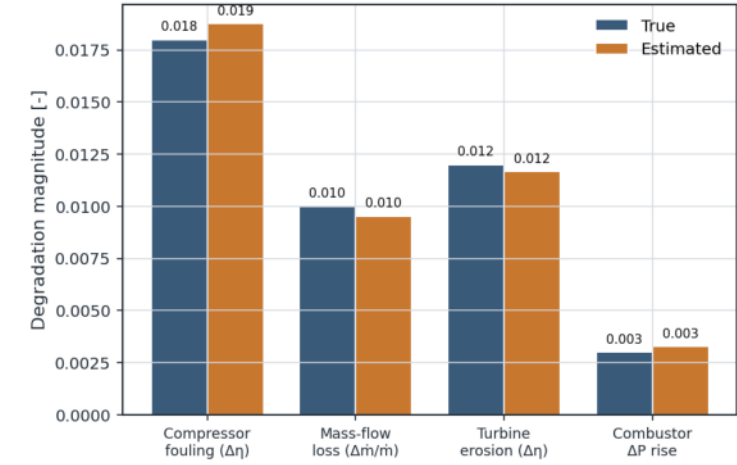


ThermoTwin-F - educational model, not validated against a specific engine

Inverse diagnostics

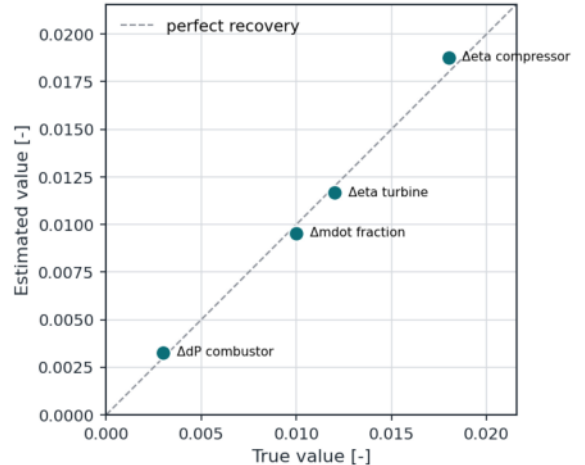
ThermoTwin-F - Inverse diagnostics

Recovered degradation parameters



ThermoTwin-F - educational model, not validated against a specific engine

Parity: estimated vs true



Assumptions & limitations

- Single-shaft open/simple Brayton cycle; no regeneration, intercooling or reheat.
- Constant specific heats by default (separate cold-air and hot-gas values). A temperature-dependent property option exists but is not the default.
- No compressor or turbine performance maps: off-design is represented by efficiency and boundary-condition changes, not by map-based matching.
- No turbine cooling-air bookkeeping; TIT is treated as the gas temperature entering the turbine.
- No detailed combustion chemistry or emissions (NOx/CO) model; combustion is a lumped energy balance with an efficiency.
- Transient model is a single lumped-capacitance metal node, not a spatially resolved thermal/FE model.
- Uncertainty and diagnostics are demonstrative: the objective function and weights are explicit choices, not a certified PTC-19.1 uncertainty budget.
- Not calibrated or validated against any specific commercial engine; not ASME PTC 22 or ISO 2314 compliant.

These limitations are intentional for a transparent teaching/portfolio tool. Each is a natural avenue for extension (maps, real-gas properties, cooling flows, emissions, FE thermal models).